



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
EMS Academic Affairs Committee & Hon. Sophia Ragaza

Power Law
Practice Problems

1. Simplify: $x^3 \times x^4$
2. Simplify: $(2x^3)^4$
3. Express with a positive exponent: x^{-5}
4. Simplify: $\frac{(x^2y^3)^3}{x^4y^7}$
5. Simplify: $(p^{-2} \times p^6)^2$
6. Simplify: $\frac{3x^2 \times (2x^3)^2}{4x^{-2}}$
7. Simplify: $\frac{y^9}{y^3}$
8. Evaluate: $5^0 \times 2^3$
9. Simplify: $(\frac{m^7}{m^2})^2$
10. Evaluate: $27^{\frac{1}{3}}$



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
EMS Academic Affairs Committee & Hon. Sophia Ragaza

Power Law
Step-by-step Solution

1. Simplify: $x^3 \times x^4$
- Rule: When multiplying terms with the same base, add the exponents:
 $a^m \times a^n = a^{m+n}$
 - Step 1: $x^3 \times x^4 = x^{3+4} = x^7$
- Answer:** x^7

2. Simplify: $(2x^3)^4$
- Rule: Apply the power to both the coefficient and the variable: $(ab)^n = a^n b^n$ and
 $(a^m)^n = a^{m \times n}$
 - Step 1: $2^4 \times (x^3)^4$
 - Step 2: $16 \times x^{3 \times 4}$
 - Step 3: $16 \times x^{12} = 16x^{12}$
- Answer:** $16x^{12}$

3. Express with a positive exponent: x^{-5}
- Rule: A negative exponent means to take the reciprocal of the base: $a^{-n} = \frac{1}{a^n}$
 - Step 1: $x^{-5} = \frac{1}{x^5}$
- Answer:** $\frac{1}{x^5}$

4. Simplify: $\frac{(x^2 y^3)^3}{x^4 y^7}$



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
EMS Academic Affairs Committee & Hon. Sophia Ragaza

- Step 1 (Numerator): Distribute the power of 3 to both terms inside the parentheses:

$$x^{2 \times 3} \times y^{3 \times 3} = x^6 y^9$$

- Step 2 (Divide): Subtract the exponents of the denominators from the numerators with

the same base: $\frac{x^6 y^9}{x^4 y^7} = x^{6-4} \times y^{9-7} = x^2 y^2$

Answer: $x^2 y^2$

5. Simplify: $(p^{-2} \times p^6)^2$

- Step 1 (Inside the parentheses): Add the exponents since the bases are the same:

$$p^{-2} \times p^6 = p^{6-2} = p^4$$

- Step 2 (Outside the parentheses): Multiply the exponents: $(p^4)^2 = p^{4 \times 2} = p^8$

Answer: p^8

6. Simplify: $\frac{3x^2 \times (2x^3)^2}{4x^{-2}}$

- Step 1 (Simplify the power): Expand $(2x^3)^2 = 4x^6$

- Step 2 (Simplify numerator): Multiply $3x^2 \times 4x^6$. Multiply the coefficients ($3 \times 4 = 12$) and add the exponents ($2+6 = 8$) to get $12x^8$ on the numerator.

- Step 3 (Divide): The expression is now $\frac{12x^8}{4x^{-2}}$. Divide the coefficients ($12 \div 4 = 3$) and subtract the exponent in the denominator from the numerator ($8 - (-2) = 8 + 2 = 10$).

Answer: $3x^{10}$

7. Simplify: $\frac{y^9}{y^3}$



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:

EMS Academic Affairs Committee & Hon. Sophia Ragaza

- Rule: When dividing terms with the same base, subtract the exponents: $\frac{a^m}{a^n} = a^{m-n}$
- Step 1: $\frac{y^9}{y^3} = y^{9-3} = y^6$

Answer: y^6

8. Evaluate: $5^0 \times 2^3$

- Rule: Any non-zero number raised to the power of 0 is 1 ($a^0 = 1$)
- Step 1: Evaluate $5^0 = 1$.
- Step 2: $5^0 \times 2^3 \Rightarrow 1 \times 8 = 8$

Answer: 8

9. Simplify: $(\frac{m^7}{m^2})^2$

- Step 1 (Inside the parentheses): Subtract the exponents: $m^{7-2} = m^5$
- Step 2 (Outside the parentheses): Multiply the exponents: $(m^5)^2 = m^{10}$

Answer: m^{10}

10. Evaluate: $27^{\frac{1}{3}}$

- Rule: A fractional exponent $\frac{1}{n}$ is equivalent to taking the n-th root: $a^{\frac{1}{n}} = \sqrt[n]{a}$
- Step 1: $27^{\frac{1}{3}} = \sqrt[3]{27} = 3$

Answer: 3